

## Project Design Phase-II

### Solution Requirements (Functional & Non-functional)

|                      |                                                    |
|----------------------|----------------------------------------------------|
| <b>Date</b>          | 17 July 2026                                       |
| <b>Team ID</b>       |                                                    |
| <b>Project Name</b>  | SB Stocks – Stock Trading Application (MERN Stack) |
| <b>Maximum Marks</b> | 4 Marks                                            |

#### Functional Requirements:

Following are the functional requirements of the SB Stocks stock trading platform.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                                                                                                                                                                                                                   |
|--------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | User Registration             | <ul style="list-style-type: none"> <li>Registration through Form (name, email, password)</li> <li>Confirmation email sent upon successful registration</li> </ul>                                                                                                                                    |
| FR-2   | User Authentication / Login   | <ul style="list-style-type: none"> <li>Login using registered email &amp; password</li> <li>JWT-based secure session management</li> <li>Logout functionality</li> </ul>                                                                                                                             |
| FR-3   | Market Dashboard              | <ul style="list-style-type: none"> <li>View real-time stock prices with auto-refresh (4-second ticker engine)</li> <li>Search stocks by symbol or company name</li> <li>View Top Gainers and Top Losers widgets</li> </ul>                                                                           |
| FR-4   | Stock Trading (Buy / Sell)    | <ul style="list-style-type: none"> <li>View price chart with Candlestick / Glowing Trend / Volume &amp; Spread views (1D, 1W, 1M, 1Y)</li> <li>Select product type (Intraday / Delivery)</li> <li>Execute Buy order at live market price</li> <li>Execute Sell order at live market price</li> </ul> |
| FR-5   | Portfolio Management          | <ul style="list-style-type: none"> <li>View current holdings (shares, avg. purchase price, current price)</li> <li>View total profit / loss per holding and overall</li> <li>View portfolio weighting via donut chart</li> </ul>                                                                     |
| FR-6   | Wallet & Fund Management      | <ul style="list-style-type: none"> <li>Deposit virtual funds via simulated funding source</li> <li>Withdraw virtual funds</li> <li>View available cash balance and net worth</li> </ul>                                                                                                              |
| FR-7   | Trading History & Audit       | <ul style="list-style-type: none"> <li>View executed order book (symbol, order type, quantity, price, status, timestamp)</li> <li>View wallet ledger of all deposit/withdrawal activity</li> </ul>                                                                                                   |
| FR-8   | User Profile Management       | <ul style="list-style-type: none"> <li>View account credentials (username, email)</li> <li>View authorization role (User / Admin)</li> <li>View net wallet balance</li> </ul>                                                                                                                        |

#### Non-functional Requirements:

Following are the non-functional requirements of the SB Stocks stock trading platform.

| FR No. | Non-Functional Requirement | Description                                                                                                                                                                                |
|--------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1  | <b>Usability</b>           | Clean, responsive dashboard UI (dark theme) with clear navigation between Dashboard, Portfolio, History, and Profile; colour-coded gains (green) and losses (red) for instant readability. |
| NFR-2  | <b>Security</b>            | JWT-based authentication, bcrypt password hashing, role-based authorization (User / Admin), input validation on all trade and wallet operations, HTTPS in production.                      |

| FR No. | Non-Functional Requirement | Description                                                                                                                                                                                    |
|--------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-3  | <b>Reliability</b>         | All trades, wallet transactions and holdings are persisted in MongoDB so data survives page refresh, logout, or server restart; failed trades are rejected before wallet/holdings are updated. |
| NFR-4  | <b>Performance</b>         | Real-time ticker engine refreshes stock prices every 4 seconds; MongoDB indexes on user and symbol fields keep portfolio and history queries fast.                                             |
| NFR-5  | <b>Availability</b>        | Frontend deployed on Vercel/Netlify and backend on Render/Railway with MongoDB Atlas, giving high uptime without manual server management.                                                     |
| NFR-6  | <b>Scalability</b>         | Stateless REST API on a 3-tier MERN architecture allows the frontend and backend to scale independently; MongoDB Atlas supports replica sets / sharding as trading volume grows.               |